

Notes: Crawford, Shcherbakov, and Shum (AER, 2019)

Quality Overprovision in Cable Television Markets

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1 Big picture

- **Question.** Does a cable monopolist underprovide or overprovide product quality relative to the social optimum when it chooses both *price* and *bundle quality* in the presence of a high-end satellite outside option?
- **Setting.** U.S. cable television markets from 1997–2006. Local cable systems choose prices and qualities for up to three bundled services, while DBS satellite competitors offer national products and prices.
- **Core challenge.** The sign of the quality distortion is theoretically ambiguous. A positive relation between cable quality and competition from satellite could come from demand conditions, technology, or a true strategic response to a high-end outside option.
- **Main conceptual contribution.** The paper moves beyond reduced-form evidence and estimates a structural demand-and-supply model with endogenous quality. It then compares observed cable choices to the allocation chosen by a social planner.
- **Main hypotheses.** The paper considers two competing ideas:
 - standard nonlinear-pricing intuition (à la Mussa-Rosen): low-quality products should be degraded;
 - high-end outside-option intuition (Champsaur-Rochet): competition from satellite can induce upward quality distortions.
- **Main empirical design.** Estimate a random-coefficients demand system plus cable firms' price and quality first-order conditions. Recover consumer preferences, marginal cost functions, and then compute counterfactual prices and qualities under social planning.
- **Main baseline result.** When the planner jointly resets both prices and qualities, the monopolist is found to set *both prices and qualities too high*. Socially optimal qualities are lower by about 23% to 55%, and socially optimal prices are lower by about 33% to 74%.
- **Key welfare result.** At the joint social optimum, consumer surplus more than doubles (about +116%) and total surplus rises by about 17%.
- **Key market-structure result.** Quality overprovision appears primarily in *multi-product* cable systems. One-product systems more often display quality degradation.
- **Key mechanism result.** The overprovision result is driven by *DBS competition*. When the satellite competitor is removed in the model, the overprovision result largely disappears and quality degradation re-emerges.
- **Economic intuition.** Cable firms use inflated quality as a business-stealing device against high-end satellite bundles. They offer too many channels and charge too much relative to what consumers would prefer under efficient pricing and quality choice.
- **Takeaway.** Market power over *quality* need not imply degradation. In markets with a high-end outside option, a monopolist may instead overprovide quality, so consumers end up buying overly large bundles at overly high prices.

2 Data and measurement (key objects)

2.1 Why U.S. cable markets are a useful laboratory

- Cable systems choose both the *price* and the *set of channels* in each bundle.
- Basic, Expanded Basic, and Digital Basic services are naturally ordered by quality: higher tiers include all lower-tier channels and more.
- Satellite TV provides a realistic high-end alternative because its bundles are typically larger and often more expensive than cable bundles.
- The period is especially informative because cable systems were largely local monopolists, but DBS entry created meaningful variation in competitive pressure.

Interpretation. This is a market where scalar product quality is a reasonable approximation, and where the theoretical distinction between a low-end and high-end outside option is economically meaningful.

2.2 Institutional background

- Cable systems operate in geographically distinct local markets.
- DBS providers (Dish and DirecTV) set products and prices nationally.
- Cable systems have substantial discretion over their channel lineups and prices, especially after the effective deregulation of most bundles by the mid-1990s.
- Before 2000, DBS competition was limited by rules restricting carriage of local broadcast channels; after local-channel carriage expands, DBS becomes a much closer substitute for cable.

Interpretation. The institutional asymmetry is central: cable firms react locally, while satellite products are effectively exogenous from the perspective of a single cable system.

2.3 Data sources and sample construction

The main data come from Warren Publishing's *Television and Cable Factbook*. Satellite product and price data are collected separately and merged in at the market-year level.

- The raw universe contains more than 11,000 cable systems per year.
- After imposing completeness restrictions, the estimation sample contains 3,931 cable systems.
- The final structural estimation sample contains 12,214 cable system-product-year observations.
- The sample covers 1997–2006 and is an unbalanced panel.

An observation is a cable-system-year with information on:

- offered bundles (up to three cable products),
- price,

- quality,
- aggregate cable market share,
- and market-level covariates.

Interpretation. The key strength of the data is that the paper observes, for each local cable system, the actual bundle menu and the underlying channel lineup, which makes endogenous quality measurement possible.

2.4 Main quality measure

The quality of a bundle is measured as the *weighted average number of television networks* in the bundle, where the weight for each channel is the national average affiliate fee paid by cable systems for that channel.

Formally, the paper treats quality as a scalar q_{gjn} for product j of provider g in market n .

Interpretation.

- Expensive channels such as ESPN, TNT, or CNN count more than inexpensive channels.
- This measure tries to capture the *economic quality* of the bundle, not just the raw channel count.

The paper also checks robustness to alternative quality measures:

- excluding free channels,
- and using the number of channels with a common low weight.

2.5 Prices and market shares

The paper observes:

- cable subscription prices for each offered tier;
- satellite prices for the competing national menus;
- aggregate cable share s_c and aggregate satellite share s_s ;
- and, for a subset of markets, product-specific cable shares.

Summary statistics:

- average aggregate cable share: 0.54;
- average aggregate satellite share: 0.15;
- average cable prices: about \$20.44 (low), \$32.77 (medium), and \$60.28 (high);
- average cable qualities: 3.09 (low), 5.76 (medium), and 9.11 (high);
- average satellite qualities are higher at every tier: 5.12, 6.48, and 10.77.

Interpretation. The fact that satellite bundles are consistently higher quality is one of the paper's most important descriptive facts. It is what makes DBS a plausible *high-end* outside option.

2.6 Other controls and cost shifters

The paper also observes market-level demand and cost shifters:

- income,
- household size,
- urbanization,
- length of the cable network,
- channel capacity,
- and other market characteristics from the 2000 U.S. Census.

Interpretation. These variables enter the demand system through heterogeneous price and quality sensitivity and enter the supply side through the marginal-cost specification.

2.7 Preliminary reduced-form evidence

Before estimating the full structural model, the paper shows:

- simple *price-per-channel* evidence of quality degradation is inconclusive;
- when DBS begins carrying local channels, cable quality tends to rise across product tiers.

Interpretation. Reduced-form patterns are suggestive, but they cannot tell us whether observed prices and qualities are above or below the efficient benchmark. That is why the structural comparison to a social planner is necessary.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and theory benchmark

The paper starts from Spence's single-product framework.

Let $P(s, q)$ be inverse demand and $C(s, q) = c(q)s$ be constant-returns cost. The monopolist chooses quantity s and quality q , with first-order conditions

$$P(s, q) + P_s(s, q)s = c(q), \tag{1}$$

$$P_q(s, q) = c_q(q). \tag{2}$$

A social planner solves

$$\max_{s, q} \left\{ \int_0^s P(s', q) ds' - c(q)s \right\},$$

which gives

$$P(s, q) = c(q), \tag{3}$$

$$\frac{1}{s} \int_0^s P_q(s', q) ds' = c_q(q). \quad (4)$$

Interpretation.

- The monopolist sets quality by equating marginal cost of quality to the *marginal consumer's* marginal valuation of quality.
- The planner instead uses the *average* marginal valuation of quality across all served consumers.
- Therefore, the sign of the distortion depends on how quality valuations vary across consumers and on the nature of the outside option.

3.2 Demand: indirect utility and heterogeneous preferences

The empirical demand model is a random-coefficients discrete-choice model.

Consumer i 's indirect utility from product j offered by provider $g \in \{c, s\}$ in market n is

$$U_{igjn} = \delta_{gjn}(p_{gjn}, q_{gjn}, \bar{Y}_{in}, \xi_g; \omega_i) + \varepsilon_{ign}, \quad (5)$$

with outside option utility normalized by ε_{i0n} .

The systematic utility is linear in price and quality:

$$\delta_{gjn}(p_{gjn}, q_{gjn}, \bar{Y}_{in}, \xi_g; \omega_i) = \alpha_{ip}(\bar{Y}_{in}) p_{gjn} + \alpha_{iq}(\bar{Y}_{in}) q_{gjn} + \xi_{gn}. \quad (6)$$

The price and quality coefficients depend on demographics:

$$\alpha_{ip} = a_{i0} + a_y I_n + a_h H_n + a_u U_n, \quad (7)$$

$$\alpha_{iq} = b_{i0} + b_y I_n + b_h H_n + b_u U_n, \quad (8)$$

and the random terms satisfy

$$\begin{pmatrix} a_{i0} \\ b_{i0} \end{pmatrix} \sim N\left(\begin{pmatrix} \bar{a}_0 \\ \bar{b}_0 \end{pmatrix}, \Sigma\right). \quad (9)$$

Interpretation.

- Consumers differ in both price sensitivity and quality valuation.
- The covariance term ρ_{ab} matters because the same consumer may have strong tastes for quality and at the same time be more or less price sensitive.
- This heterogeneity is essential for evaluating whether quality distortions come from preference curvature or from the competitive environment.

3.3 Provider choice probabilities and shares

Given the Type-I extreme value assumption, the probability that consumer type i chooses provider g in market n is

$$P_{gn}(\omega_i) = \frac{\exp(\delta_{gn})}{1 + \exp(\delta_{cn}) + \exp(\delta_{sn})}, \quad (10)$$

and the aggregate share is

$$s_{gn} = \int P_{gn}(\omega_i) dF(\omega_i). \quad (11)$$

For markets where product-specific cable shares are observed, the paper also uses the conditional share equation

$$s_{j|g,n} = \frac{\int \mathbf{1}(j = \arg \max_{j' \in \mathcal{J}_{gn}} \delta_{gj'n}) P_{gn}(\omega_i) dF(\omega_i)}{\int P_{gn}(\omega_i) dF(\omega_i)}. \quad (12)$$

Interpretation. Aggregate cable versus satellite choice identifies substitution across providers, while the product-specific conditional shares help identify substitution within the cable menu.

3.4 Supply: cable firms choose price and quality

Cable firms maximize variable profit product by product:

$$\max_{\{p_{cjn}, q_{cjn}\}_{j \in \mathcal{J}_{cn}}} \sum_{j \in \mathcal{J}_{cn}} s_{cjn}(\cdot) [p_{cjn} - mc(q_{cjn}, Z_{cn}; \theta_s)]. \quad (13)$$

The satellite competitor is treated as nonstrategic from the perspective of a local cable operator.

Interpretation.

- Cable chooses a local menu strategically.
- DBS products and prices evolve nationally and are taken as given by the local cable system.
- This assumption fits the institutional setting because one local cable firm's decisions are too small to trigger a national DBS repricing.

3.5 Supply first-order conditions

For each cable product j , the first-order condition for price is

$$s_{cjn} + \sum_{r \in \mathcal{J}_{cn}} (p_{crn} - mc_{crn}) \frac{\partial s_{crn}}{\partial p_{cjn}} = 0, \quad (14)$$

and the first-order condition for quality is

$$-\frac{\partial mc_{cjn}}{\partial q_{cjn}} s_{cjn} + \sum_{r \in \mathcal{J}_{cn}} (p_{crn} - mc_{crn}) \frac{\partial s_{crn}}{\partial q_{cjn}} = 0. \quad (15)$$

Interpretation.

- The pricing equation is the standard markup condition.
- The quality equation says that cable raises quality until the demand benefit from higher quality equals the marginal cost of supplying it.
- These first-order conditions let the authors back out the marginal cost level and its slope at observed qualities.

3.6 Marginal cost function

To compute counterfactuals away from the observed quality points, the paper imposes

$$mc(q_{cjn}, Z_{cn}; \theta_s) = \exp(c_{0jn} + c_{1jn}q_{cjn}). \quad (16)$$

Then the implied local cost coefficients are projected on observed cost shifters:

$$\hat{c}_{0jn} = Z_{cjn}\theta_{s0} + \nu_{0cjn}, \quad (17)$$

$$\hat{c}_{1jn} = Z_{cjn}\theta_{s1} + \nu_{1cjn}. \quad (18)$$

Interpretation.

- The exponential form ensures marginal cost is positive and increasing in quality.
- The paper then uses monotone Hermite splines between observed quality points to build flexible cost curves for counterfactual simulations.

3.7 Why the number of products is treated as exogenous

The paper does *not* model the menu-size choice. Instead, it takes the number of offered cable products as given.

The justification is institutional:

- the number of products depends heavily on installed technology and channel capacity;
- those menu-size choices are slower-moving than annual price and quality decisions.

Interpretation. The model is best understood as a short-run quality-and-price-choice model conditional on the cable system's existing menu size.

3.8 Identification and estimation

The authors solve for cable and satellite demand shocks (ξ_{cn}, ξ_{sn}) using a Berry-Levinsohn-Pakes-style inversion based on observed shares. They then use:

- moments from demand shocks,
- moments from supply-side cost shocks,
- and moments from measurement error in conditional cable product shares.

Structural parameters are estimated by GMM with an optimal weighting matrix.

Interpretation. The model handles the endogeneity of both *price* and *quality*. This is important because quality is a choice variable and is likely to be correlated with unobserved demand conditions.

3.9 Instrument set

The baseline instruments are built from ownership and technology features:

- average prices and qualities of *other systems in the same MSO*;
- the number of MSO subscribers;
- average MSO channel capacity;
- own channel capacity;
- total length of own coaxial lines.

The logic is:

- common MSO ownership creates correlated prices, qualities, and marginal costs across local systems;
- but local market-specific demand shocks need not be correlated across those systems because cable was not nationally advertised in the same way.

Interpretation. The identification strategy tries to move price and quality with cost-side and organization-side shifters, while leaving local demand shocks behind.

3.10 Alternative specifications

The paper reports:

- an OLS-type specification where price and quality instrument for themselves;
- a minimal-IV specification using only MSO-based instruments;
- alternative measures of quality.

The qualitative results remain intact, especially the finding that joint optimization implies quality overprovision.

Interpretation. This is reassuring because the paper's main message would be fragile if it depended completely on one instrument set or one exact quality measure.

3.11 Social planner problem

Consumer surplus for market n is

$$E(CS_n) = - \int \frac{1}{\alpha_{ip}} \ln \left(1 + \sum_{g=c,s} \exp(\delta_{gjn}(\cdot)) \right) dF(\omega_i). \quad (19)$$

The social planner chooses cable prices and qualities to satisfy

$$\frac{\partial \Pi(\cdot)}{\partial p_{cjn}} + \frac{\partial E(CS_n)}{\partial p_{cjn}} = 0, \quad (20)$$

$$\frac{\partial \Pi(\cdot)}{\partial q_{cjn}} + \frac{\partial E(CS_n)}{\partial q_{cjn}} = 0. \quad (21)$$

Interpretation.

- The planner internalizes both consumer surplus and firm profit.
- Comparing the observed monopoly allocation to the planner allocation delivers the paper's central welfare and distortion results.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline results

There is no external quasi-experiment in the main structural estimation. Credibility comes from combining:

- observed local cable menus and prices,
- exogenous national DBS menus and prices,
- product-specific share information where available,
- demand-side instruments,
- supply-side cost shifters,
- and strong institutional restrictions on how the industry works.

Interpretation. The paper identifies the model by exploiting differences across local cable systems in costs, channel capacity, ownership structure, and competitive exposure to DBS.

4.2 Why reverse causality is limited on the DBS side

A potential concern is that DBS could respond strategically to cable quality. The paper argues this is second-order because:

- DBS products and prices were set nationally;
- local cable firms made decisions at a much finer geographic level;
- one local cable system is too small to induce a national DBS menu adjustment.

Interpretation. This supports treating DBS as nonstrategic in the local cable firm's problem.

4.3 How the paper handles endogenous quality

Quality is clearly endogenous: firms choose more channels when demand is strong. The model addresses this by:

- solving for provider-specific demand shocks ξ_{gn} ,
- instrumenting prices and qualities with MSO and capacity variables,
- and estimating demand and supply jointly.

Interpretation. This is one of the paper’s main econometric contributions: it is not just estimating demand *given* quality, but a market-equilibrium model where quality itself is chosen.

4.4 Why the quality overprovision result is not just an artifact of random-coefficient covariance

The estimated correlation between the random coefficients on price and quality is negative:

$$\rho_{ab} \approx -0.481$$

in the preferred demand specification.

Because covariance parameters can be weakly identified in random-coefficients models, the paper varies ρ over a wide range in the simulations. The main message survives:

- with DBS present, quality overprovision remains across a wide range of ρ ;
- without DBS, the model reverts largely to quality degradation.

Interpretation. This matters because the headline finding would be much less convincing if it only appeared for one knife-edge covariance estimate.

4.5 What the paper can distinguish

The paper can distinguish fairly clearly between:

- *quality degradation at fixed prices,*
- *price distortion holding quality fixed,*
- and *joint distortion in price and quality.*

It can also distinguish between:

- a world with DBS competition,
- and a counterfactual world without DBS competition.

Interpretation. This is why the paper can show the subtle but important point that cable *underprovides* quality if prices are held fixed, yet *overprovides* quality once prices are allowed to adjust jointly.

4.6 What the paper cannot fully distinguish

The paper is more limited in separating:

- preference-based overprovision à la Spence,
- from high-end-outside-option overprovision à la Champsaur-Rochet,

because both can in principle generate upward distortions.

Still, the paper shows that:

- overprovision is much stronger in multi-product systems;
- and it largely disappears when DBS is removed.

Interpretation. That pattern points strongly toward the *high-end outside option* mechanism as the dominant explanation in this market.

4.7 Relation to the literature

The paper is closest to:

- Spence (single-product monopoly quality),
- Mussa-Rosen and the nonlinear-pricing/screening literature,
- Champsaur-Rochet on high-end outside options,
- and empirical work on cable quality and price choice such as Crawford and Shum and Chu.

Its key novelty is empirical:

- most prior theory and evidence stressed *degradation*;
- this paper finds economically important *overprovision*.

5 Tables and figures

5.1 Figure 1: market shares and the structure of competition

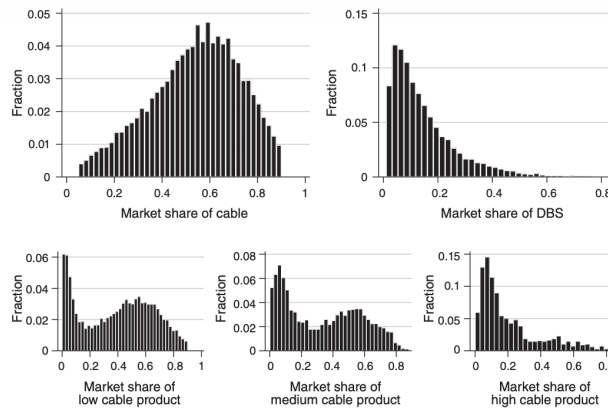


FIGURE 1. HISTOGRAMS FOR CABLE AND SATELLITE MARKET SHARES

Figure 1: Figure 1 from the paper: cable and satellite market-share distributions.

Read:

- Aggregate cable shares are centered at relatively high levels.
- Satellite shares are lower on average but broadly distributed.

- Product-specific cable shares fall with tier quality.

Economic message. Cable is still the dominant platform, but DBS is large enough to be a meaningful high-end competitive threat rather than a negligible outside option.

5.2 Table 2 and Table 4: data summary and reduced-form evidence

TABLE 2—DATA SUMMARY STATISTICS, 1997–2006

	Mean	SD	Min	Max
<i>Periods and products</i>				
Time periods, years	2.1	1.2	1	9
Cable products	1.5	0.7	1	3
Satellite products	5.2	1.4	3	6
<i>Market shares</i>				
s^c	0.54	0.19	0.05	0.90
s^s	0.15	0.12	0.01	0.88
<i>Prices</i>				
<i>Cable</i>				
p_{low}^c	\$20.44	\$7.71	\$2.68	\$81.86
p_{med}^c	\$32.77	\$14.59	\$7.88	\$136.30
p_{high}^c	\$60.28	\$23.79	\$16.90	\$291.08
<i>Satellite</i>				
p_{low}^s	\$21.69	\$9.33	\$14.44	\$39.24
p_{med}^s	\$27.15	\$9.06	\$19.26	\$43.61
p_{high}^s	\$45.95	\$22.94	\$28.89	\$87.22
<i>Quality</i>				
<i>Cable</i>				
q_{low}^c	3.09	1.81	0.30	13.13
q_{med}^c	5.76	2.63	0.71	16.86
q_{high}^c	9.11	3.45	2.34	19.02
<i>Satellite</i>				
q_{low}^s	5.12	3.49	1.78	11.73
q_{med}^s	6.48	3.16	3.30	12.67
q_{high}^s	10.77	6.51	5.65	27.88
<i>Other variables</i>				
Length of cable network, miles	0.160	0.560	0.000	17.690
Channel capacity	44.9	20.2	5	542
<i>Before 2002</i>				
p_{tier3}^c	\$35.98	\$4.26	\$32.75	\$43.80
p_{tier4}^c	\$41.10	\$2.92	\$38.53	\$46.59
p_{tier5}^c	\$54.67	\$12.65	\$46.23	\$77.35
q_{tier3}^c	7.42	1.78	5.93	10.44
q_{tier4}^c	8.01	1.75	6.53	10.96
q_{tier5}^c	9.45	3.49	6.96	15.44

Notes: This table reports summary statistics for our estimation sample. Cable information is drawn from Warren Publishing's *Television and Cable Factbook*; satellite information was collected by hand. There are as many as six satellite products until 2001 and only three afterward. Market shares reported in the table are aggregated across all products offered on each platform (cable, satellite). Prices are in 1997 dollars. Low (or the only one if 1 product), Medium (second highest across having 2+ products), and High (third among 3-product markets) quality or price products are indexed by $\{0, 1, 2\}$ in the modeling section. The quality of a product is the weighted average of the number of television networks offered on that product, with weights given by the national average input cost (affiliate fee) paid by cable systems for that network from SNL Kagan. Length of cable network measures the size of the physical cable network, while channel capacity measures the maximum number of channels a given cable system can broadcast using its current physical network and compression technology. These data are augmented with market-level data on demographics from 2000 US Census.

TABLE 4—CABLE QUALITY CHOICE AND DBS CARRIAGE OF LOCAL CHANNELS

Variable	DBS carriage of local channels	No DBS carriage of local channels	Difference
1-product, q_{c0}	4.09	3.87	0.22
2-product, q_{c0}	4.51	3.97	0.54
2-product, q_{c1}	6.38	5.94	0.44
3-product, q_{c0}	3.01	2.44	0.57
3-product, q_{c1}	8.95	7.92	1.03
3-product, q_{c2}	10.85	9.53	1.32

Read:

- Average cable share is 0.54, average satellite share is 0.15.
- Cable offers between one and three products; satellite offers between three and six, then three from 2002 onward.
- Satellite bundles are higher quality than cable bundles at every tier.
- When DBS carries local channels, cable quality is higher across market/product types.

Economic message. The descriptive evidence is already consistent with the idea that DBS is a high-end outside option and that cable firms respond by raising quality.

5.3 Table 5 and Figure 2: demand estimates

Read:

TABLE 5—ESTIMATION RESULTS

	Coefficient	SE	Coefficient	SE
	(1)		(2)	
<i>Price coefficient, α_{ip}</i>				
Mean	−0.673	(0.008)	−0.682	(0.035)
Income			−0.418	(0.147)
Household size			0.396	(0.271)
Urban			−0.173	(0.177)
Sigma	0.102	(0.018)	0.122	(0.018)
<i>Price coefficient, α_{iq}</i>				
Mean	1.108	(0.121)	1.225	(0.241)
Income			−0.337	(0.777)
Household size			0.221	(1.395)
Urban			0.331	(0.917)
Sigma	0.310	(0.020)	0.266	(0.048)
ρ_{ab}	−0.545	(0.128)	−0.481	(0.049)
Demand <i>t</i> -dummies	Yes		Yes	
Supply <i>t</i> -dummies	Yes		Yes	

Notes: Reported are estimation results for key parameters from our structural model of demand, price, and quality choice. There are 12,214 observations, where an observation is a cable system-product-year. Full estimation results can be found in Appendix C. Specification (1) does not include market demographics in preferences, while specification (2) does. Standard errors are clustered by market and listed in parentheses. Instruments include all those described in Section IVC; results using each variable as its own instrument (OLS-type) and a minimal set of instruments based on ownership (MSO) measures (min-IV) are presented in Appendix B. First-stage regressions are reported in Appendix D.

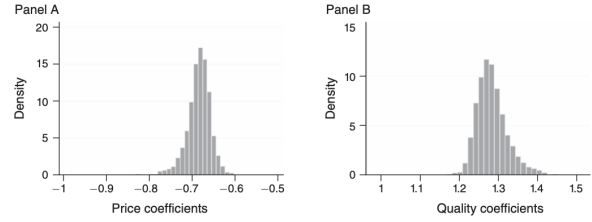


FIGURE 2. DISTRIBUTIONS OF MEAN PRICE AND QUALITY PARAMETERS ACROSS MARKETS

Note: Reported are the estimated distributions of mean price and quality parameters across markets implied by the parameter estimates reported in Table 5.

- Preferred specification (column 2) gives mean price coefficient -0.682 and mean quality coefficient 1.225 .
- The standard deviation of quality tastes is much larger than the standard deviation of price tastes.
- The estimated covariance between price and quality taste shifters is negative, $\rho_{ab} = -0.481$.
- The distribution of mean price and quality parameters across markets is fairly tight around those averages.

Economic message. Demand is reasonably elastic, consumers differ substantially in their willingness to pay for quality, and the model leaves room for both quality degradation and quality overprovision depending on the competitive environment.

5.4 Table 6 and Figure 3: supply-side estimates and marginal costs

Read:

- One-product systems have the highest and steepest marginal cost curves.
- Two-product systems are cheaper than one-product systems at comparable quality levels.
- Three-product systems have the flattest cost curves at higher quality levels.

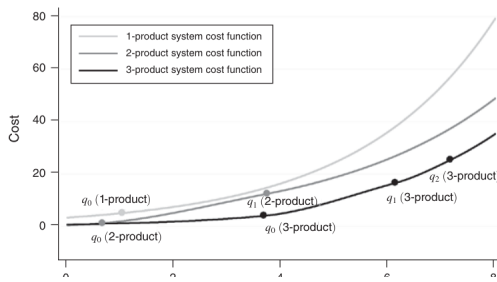
Selected numbers:

- one-product mean marginal cost at observed quality: 16.81;
- two-product means: 13.36 for the low product and 25.47 for the high product;
- three-product means: 12.88, 28.61, and 47.80 for low, medium, and high products.

Economic message. Multi-product cable firms are more cost-efficient at supplying high quality, which helps explain why quality overprovision is concentrated in two- and three-product markets.

TABLE 6—SUMMARY STATISTICS FOR THE SUPPLY-SIDE ESTIMATES

	Mean	Median	Min	Max	SD
System type: 1-product					
mc_{c0}	16.81	17.15	0.03	46.79	5.79
$\partial mc_{c0}/\partial q_{c0}$	1.90	1.90	1.03	2.96	0.13
System type: 2-product					
mc_{c0}	13.36	11.44	0.00	70.42	9.42
mc_{c1}	25.47	22.53	1.50	96.06	11.93
$\partial mc_{c0}/\partial q_{c0}$	1.79	1.84	0.07	2.41	0.25
$\partial mc_{c1}/\partial q_{c1}$	13.49	7.41	0.66	114.75	13.71
System type: 3-product					
mc_{c0}	12.88	12.16	0.06	60.46	8.58
mc_{c1}	28.61	26.31	2.20	99.36	12.97
mc_{c2}	47.80	45.12	11.46	119.67	17.43
$\partial mc_{c0}/\partial q_{c0}$	1.77	1.83	0.12	2.40	0.26
$\partial mc_{c1}/\partial q_{c1}$	4.03	3.64	0.27	14.31	1.89
$\partial mc_{c2}/\partial q_{c2}$	20.79	16.73	2.04	129.47	14.74



5.5 Table 7: welfare decomposition

Read:

- *Column 1, observed prices with planner quality:* low-quality products are degraded by 55%; medium and high products by about 7%; consumer surplus rises by 43%; total surplus rises by 10%.
- *Column 2, planner price with planner quality fixed:* prices fall by 33%, 59%, and 74% across low, medium, and high products; consumer surplus rises by 54%; total surplus rises by 7%.
- *Column 3, full planner allocation:* socially optimal qualities are lower by 23%, 37%, and 55%; consumer surplus rises by 116%; total surplus rises by 17%.

Economic message. The really important distortion is *joint*. If one holds price fixed, cable appears to underprovide quality. But once both price and quality are allowed to move, the cable monopolist is revealed to be choosing overly large bundles at overly high prices.

5.6 Figure 4: where overprovision appears

Read:

- In one-product markets, the mass is mostly on the *degradation* side.
- In two-product markets, the mass shifts to the *overprovision* side.
- In three-product markets, overprovision is even more dominant.

The paper reports:

TABLE 7—WELFARE EFFECTS OF MARKET POWER OVER QUALITY (AND PRICE)

	Market power over quality		Market power over price		Total welfare effect	
	(p^{Obs}, q^{Obs}) vs. (p^{Obs}, q^{SP})		(p^{Obs}, q^{SP}) vs. (p^{SP}, q^{SP})		(p^{Obs}, q^{Obs}) vs. (p^{SP}, q^{SP})	
	Mean	SD	Mean	SD	Mean	SD
	(1)		(2)		(3)	
Prices						
Low-quality products	–	–	–0.330	0.180	–0.330	0.180
Medium-quality products	–	–	–0.590	0.220	–0.590	0.220
High-quality products	–	–	–0.740	0.130	–0.740	0.130
Qualities						
Low-quality products	0.550	0.720	–	–	–0.230	0.910
Medium-quality products	0.070	0.110	–	–	–0.370	0.410
High-quality products	0.070	0.040	–	–	–0.550	0.260
Welfare						
Consumer surplus	0.430	0.290	0.540	0.420	1.160	0.520
Profit	–0.330	0.240	–1.000	–	–1.000	–
Total surplus	0.100	0.060	0.070	0.050	0.170	0.070

Note: Numbers in this table are proportional changes, relative to the baseline scenarios (at observed prices and qualities).

- only about 20% of one-product markets exhibit overprovision;
- about 85% of two-product markets do;
- about 95% of three-product markets do.

Economic message. Overprovision is mainly a multi-product phenomenon. That pattern fits the story that multi-tier cable systems strategically mimic or chase DBS quality where they have the cost structure to do so.

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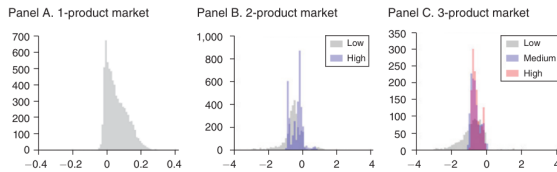


FIGURE 4. QUALITY INFLATION FOR EACH PRODUCT BY TYPE OF CABLE SYSTEM

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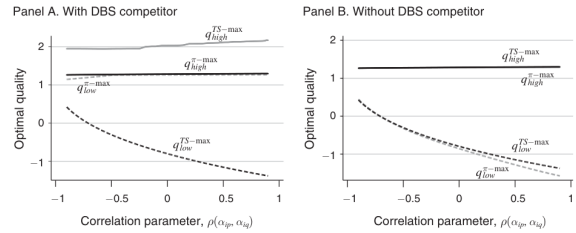


FIGURE 5

Note: Benchmark two-product market (Evansville, IL, 1997): profit and TS maximizing cable quality for monopolist with (panel A) and without (panel B) the presence of DBS competitor.

and without competition from satellite providers. With these quantities we can now

5.7 Figure 5: with and without DBS competition

Read:

- In the benchmark two-product market (Evansville, IL, 1997), quality overprovision remains for a wide range of ρ when DBS is present.
- When DBS is removed, the monopolist's chosen quality falls dramatically and moves much closer to the planner's choice.
- For the high-quality product, monopoly quality with DBS is roughly 80% above the planner's quality over a broad parameter range.

Economic message. This figure is the clearest visual evidence for the paper's mechanism: DBS competition is what flips the sign from degradation to overprovision.

5.8 Figure 7: no DBS means degradation

Read:

- Once DBS is removed from the counterfactual, the histograms for 1-, 2-, and 3-product systems shift overwhelmingly into the degradation region.
- This is true even in markets where the paper had found strong overprovision with DBS present.

Economic message. The overprovision result is not a generic feature of cable monopoly. It is a feature of monopoly *facing a high-end outside option*.

TABLE 7—WELFARE EFFECTS OF MARKET POWER OVER QUALITY (AND PRICE)

	Market power over quality (p^{Obs}, q^{Obs}) vs. (p^{SP}, q^{SP})		Market power over price (p^{Obs}, q^{SP}) vs. (p^{SP}, q^{SP})		Total welfare effect (p^{Obs}, q^{Obs}) vs. (p^{SP}, q^{SP})	
	Mean	SD	Mean	SD	Mean	SD
	(1)		(2)		(3)	
<i>Prices</i>						
Low-quality products	—	—	−0.330	0.180	−0.330	0.180
Medium-quality products	—	—	−0.590	0.220	−0.590	0.220
High-quality products	—	—	−0.740	0.130	−0.740	0.130
<i>Qualities</i>						
Low-quality products	0.550	0.720	—	—	−0.230	0.910
Medium-quality products	0.070	0.110	—	—	−0.370	0.410
High-quality products	0.070	0.040	—	—	−0.550	0.260
<i>Welfare</i>						
Consumer surplus	0.430	0.290	0.540	0.420	1.160	0.520
Profit	−0.330	0.240	−1.000	—	−1.000	—
Total surplus	0.100	0.060	0.070	0.050	0.170	0.070

Note: Numbers in this table are proportional changes, relative to the baseline scenarios (at observed prices and qualities).

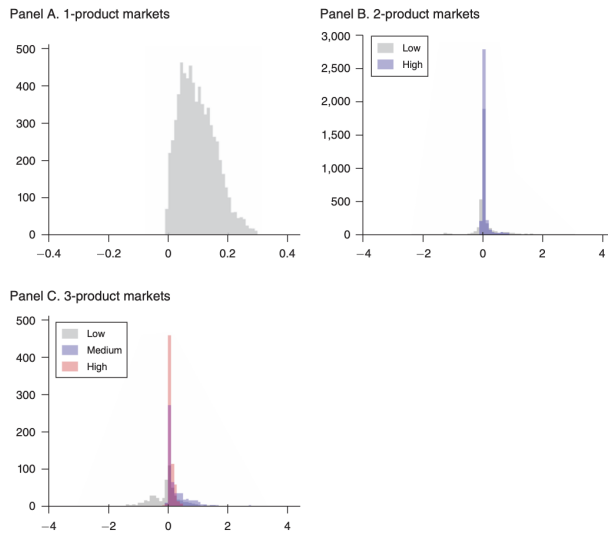


FIGURE 7. QUALITY INFLATION FOR EACH PRODUCT BY TYPE OF CABLE SYSTEM: SOCIALLY OPTIMAL AND PROFIT MAXIMIZING IF DBS IS OUT

Note: Quality inflation is defined in percentage terms as the difference between socially optimal and profit-maximizing quality level when DBS is eliminated from the market.

5.9 Other important quantitative results not shown above

Demand and markups.

- Average own-price elasticity is about -5.15 in one-product markets and -5.26 to -5.28 in multi-product markets.
- Estimated price markups average about 26.4% in one-product markets, 32.0% and 22.6% in two-product markets (low/high), and 33.9%, 17.4%, and 21.0% in three-product markets (low/medium/high).

Economic message. Cable has meaningful market power over price, but the paper's punchline is that this interacts with market power over quality in a nonstandard way.

6 Short summary

- The paper asks whether local cable monopolists distort quality upward or downward relative to the efficient benchmark.
- It estimates a structural model of demand and cable firms' price-and-quality choices using U.S. cable markets from 1997–2006.
- The key institutional fact is that DBS satellite bundles provide a high-end outside option.
- At observed prices, monopoly quality looks degraded, especially for low-quality products.
- But when the social planner jointly resets both prices and qualities, the monopoly allocation is revealed to have *too much quality and too high prices*.
- Consumers would prefer smaller bundles at lower prices, with consumer surplus more than doubling at the social optimum.
- The result is strongest in multi-product cable markets and largely disappears when DBS competition is removed.
- The paper's main contribution is to show empirically that market power over quality can generate *overprovision*, not only degradation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that, in U.S. cable television markets, a monopolist choosing both prices and qualities tends to set *both too high* relative to the first-best social optimum. The size of the distortion is economically large and robust across many specifications.

7.2 Economic intuition

The intuition is not that firms love quality for its own sake. Rather, quality is used strategically against satellite competition. Because DBS offers nationally fixed, high-quality bundles, a local cable monopolist has an incentive to inflate bundle quality to steal business, even though consumers would be better off with smaller bundles at lower prices.

7.3 Takeaway

The paper changes the usual lesson from the quality-distortion literature. Monopoly need not mean “too little quality.” In the presence of a high-end outside option, monopoly can mean *too much quality bundled with too high a price*. For policy, that means improving consumer welfare may require not just lower prices, but also *smaller and cheaper product designs*.